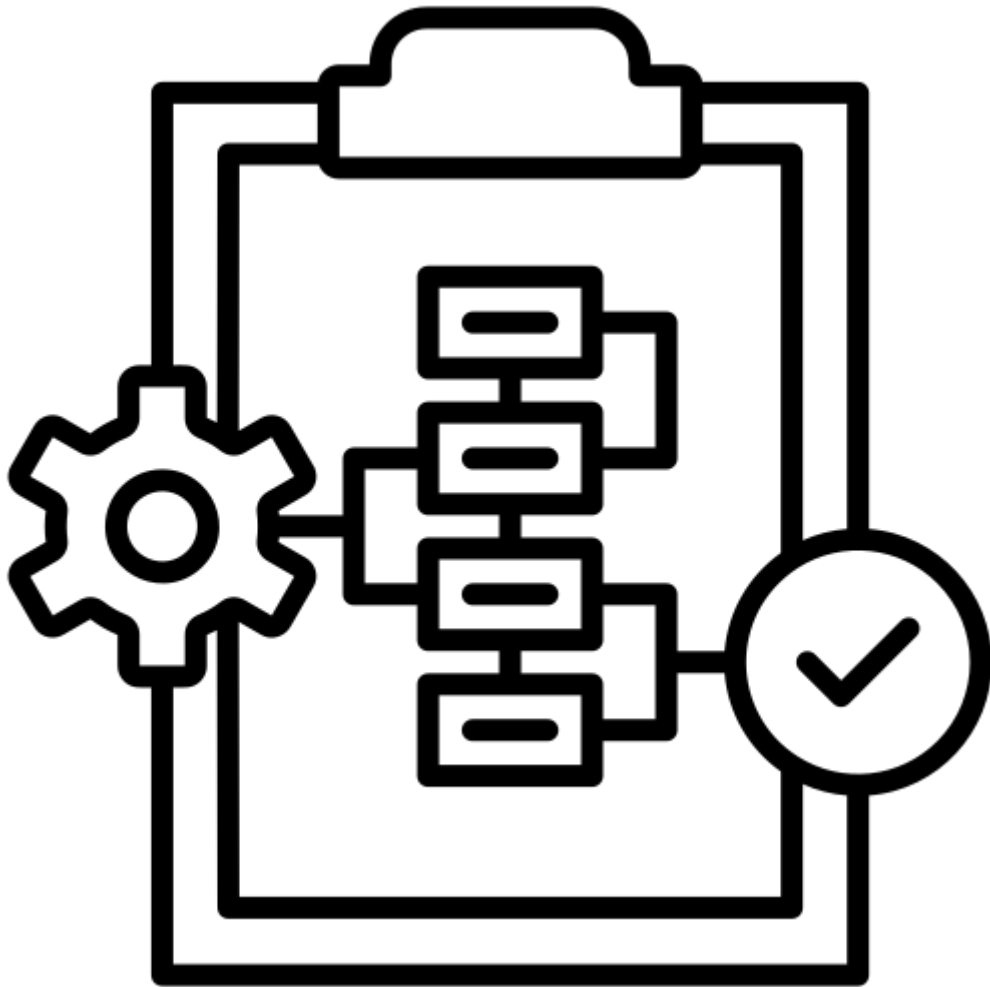


Lesson 4: Pre-Deployment Procedures



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Pre-Deployment Scenario Discussion

Objective:

This activity introduces learners to the essential processes involved in setting up and securing a dedicated web server before deploying an e-commerce platform. Through critical reflection, students will explore the importance of these processes and analyze potential risks if these steps are not followed.

Scenario:

A corporation is preparing to launch its new e-commerce platform and must set up a dedicated web server to host the application. The IT team is tasked with ensuring that the server is secure, properly configured, and capable of handling the anticipated volume of client traffic

Instructions:

Read the scenario carefully and reflect on the following guide questions:

1. **Pre-Deployment Preparation: (Put Textbox per questions in website)**
 - What essential processes should the IT team follow to ensure the server is ready for launch?

- Consider aspects such as server configuration, security protocols, performance optimization, and backup systems.

2. Importance of Pre-Deployment Processes:

- Why are these processes critical for maintaining operational awareness and ensuring security?
- How do these steps contribute to the platform's reliability and customer trust?

3. Risk Assessment:

- What potential risks could arise if these procedures are not followed?
- Reflect on possible security breaches, performance issues, and system failures that may affect business operations.



What I Need to Know?

Objectives:

At the end of the lesson, the students should be able to:

1. **Describe** essential pre-deployment processes and their importance for achieving operational awareness.
2. **Perform** proper procedures to prepare the hardware and software for deployment, exhibiting the right methods and safety precautions.
3. **Express** the importance of pre-deployment practices and how they affect work effectiveness and teamwork through a personal reflection.

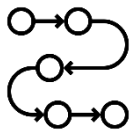


LET'S LEARN:

Introduction to Pre-Deployment Procedures

Pre-deployment procedures in the context of setting-up computer servers refer to the critical steps performed before deploying and configuring a server. These procedures ensure that the server environment is properly prepared, configured, and tested to meet operational requirements. Whether setting up a web server, database server, or application server, following a structured pre-deployment plan minimizes risks and ensures smooth operation.

Server deployment involves installing necessary hardware and software components, configuring network settings, and ensuring security protocols are in place. Proper pre-deployment practices prevent errors that could disrupt system performance, compromise security, or result in service downtime. Additionally, pre-deployment procedures ensure that the server meets the needs of the organization, supports scalability, and complies with relevant standards and protocols.



Essential Pre-Deployment Procedures

To achieve a successful server setup, it is essential to follow these pre-deployment steps:

1. Requirement Analysis and Planning

- Define the server's purpose, such as web hosting, database management, or file sharing.
- Identify hardware and software specifications based on workload, storage, and scalability needs.

2. Hardware and Software Preparation

- Verify that server hardware meets performance and capacity requirements.
- Install and configure the operating system (OS) and essential server applications.
- Ensure that all required software licenses and subscriptions are valid.

3. Network Configuration

- Assign static IP addresses, configure DNS settings, and establish appropriate firewall rules.
- Ensure proper port configuration to allow necessary communication between server and clients.

- Configure Virtual LANs (VLANs) if multiple subnets or security zones are required.

4. Security Assessment and Hardening

- Implement security protocols, including firewalls, intrusion detection systems, and encryption.
- Configure strong password policies and multi-factor authentication (MFA).
- Regularly update and patch the server's OS and applications to address vulnerabilities.

5. Backup and Disaster Recovery Plan

- Establish regular backup routines to prevent data loss.
- Define and test a rollback or disaster recovery plan to minimize downtime in case of failure.

6. Testing and Quality Assurance

- Perform functional, performance, and security testing to ensure the server meets operational requirements.
- Conduct user acceptance testing (UAT) to validate the system's functionality.
- Stress tests the server to assess its capacity and reliability under peak loads.

7. Documentation and Training

- Create detailed documentation of server configurations, network settings, and backup procedures.
- Provide training for system administrators and support staff on managing and troubleshooting the server.



Potential Risks of Inadequate Pre-Deployment Procedures

Failure to follow thorough pre-deployment procedures can lead to significant risks and operational challenges. Some potential risks include:

1. Server Downtime and System Failures

- Improper configuration may cause the server to crash or become unresponsive.
- Inadequate performance tuning can result in slow response times and service outages.

2. Security Vulnerabilities and Data Breaches

- Weak security measures can expose the system to hacking, malware, and unauthorized access.
- Lack of regular updates and patching increases the risk of exploitation by attackers.

3. Loss of Critical Data

- Absence of backup procedures can lead to permanent data loss.
- Insufficient disaster recovery planning may extend downtime after a failure.

4. Compliance and Legal Issues

- Failure to comply with industry regulations, such as GDPR or HIPAA, may result in legal consequences.
- Non-compliance with security standards may lead to audits, penalties, and reputational damage.

5. Increased Operational Costs

- Fixing deployment errors post-deployment may consume significant resources and time.
- Extended downtimes may lead to financial losses and customer dissatisfaction.

6. Poor Scalability and Performance

- Lack of performance testing may lead to server bottlenecks as user demand grows.
- Misconfigured resources may prevent the server from scaling efficiently.



Let's Apply!

Activity: Understanding Pre-Deployment Procedures for Setting Up Computer Servers

Objective:

To familiarize students with the essential hardware and software pre-deployment procedures for setting up computer servers and ensure they understand the importance of backups.

Scenario

You are an IT technician preparing a server room for a university's new student portal. Your task is to ensure all hardware and software components meet performance, security, and reliability standards before deployment.

Task Instructions

1. Hardware Inspection & Testing

Objective: Validate hardware reliability and performance.

1. Physical Inspection

- Visually inspect servers, workstations, and storage devices for physical damage (e.g., bent pins, cracked components).
- Document any defects and report them for replacement or repair.

2. Performance Stress Testing

- **Tools:** Use Prime95 (CPU), MemTest86 (RAM), and FurMark (GPU).
- **Procedure:**
 - Run each tool for 1–2 hours to simulate heavy workloads.
 - Monitor temperatures using HWMonitor to identify overheating risks.
 - Record performance metrics (e.g., CPU throttling, memory errors).

3. Bottleneck Analysis

- Identify components causing slowdowns during stress tests (e.g., insufficient RAM, disk I/O limitations).
- Recommend hardware upgrades if bottlenecks exceed 20% performance loss.

2. Software Preparation & Validation

Objective: Ensure software stability, compatibility, and security.

1. Compatibility Verification

- Confirm the operating system (e.g., Windows Server 2022) meets the software's minimum requirements.
- Install all OS updates and patches before deploying applications.

2. Dependency Installation

- Install required libraries/frameworks (e.g., .NET Core for web apps).

- Configure environment variables and runtime settings (e.g., Java heap space).

3. Functional and Security Testing

- **Functional Tests:**

- Simulate user workflows (e.g., 100 concurrent logins) to verify feature stability.
- Use load-testing tools like JMeter to measure response times under stress.

- **Security Tests:**

- Run vulnerability scans with OWASP ZAP to detect misconfigurations or exploits.
- Address critical vulnerabilities (e.g., unpatched CVEs) before deployment.

4. Backup Protocols

- Create full system backups using Veeam or Windows Server Backup.
- Perform a test restore to ensure backup integrity.

Reflection Questions (**Again put textboxes for this**)

1. Why is stress testing critical for anticipating real-world server demands?
2. How might unpatched software vulnerabilities impact a university's data security?
3. What steps would you take if a backup restoration failed during deployment?

The Criteria for the Performance Task

Category	Excellent (10)	Good (8)	Satisfactory (6)	Needs Improvement (4)	Incomplete (0)
Hardware Inspection & Testing	Thorough inspection with detailed documentation of defects and performance metrics. Bottleneck analysis leads to actionable upgrade recommendations.	Minor omissions in documentation, but thorough analysis. Bottleneck identification is mostly accurate.	Inspection is basic, some performance metrics missing. Bottleneck analysis is incomplete.	Major errors in inspection or testing; no actionable recommendations.	No attempt or incomplete.
Software Preparation & Validation	Software fully compatible with OS and hardware. All dependencies installed and configured correctly. Security tests identify no critical vulnerabilities.	Minor compatibility issues, but software functions. Most dependencies correctly installed. Some minor security vulnerabilities noted.	Software partially compatible, some dependencies missing. Security tests reveal moderate vulnerabilities.	Major compatibility issues or missing dependencies. Critical security vulnerabilities remain unaddressed.	No attempt or incorrect configuration.
Backup Protocols	Full system backups created and successfully	Backups mostly complete, minor	Basic backups created, but	Major errors in backup creation or restore process.	No backups created or attempted.

	restored. Backup integrity verified.	issues with restore process.	restore process not fully tested.		
Documentation & Evidence	Detailed step-by-step guide with screenshots for all steps. Clear evidence of successful testing.	Mostly detailed documentation, some screenshots missing. Evidence of testing is mostly complete.	Basic documentation with some missing details. Limited evidence of testing.	Poorly structured documentation with critical steps missing. Minimal evidence of testing.	No documentation or extremely incomplete.
Hardware Inspection & Testing	Thorough inspection with detailed documentation of defects and performance metrics. Bottleneck analysis leads to actionable upgrade recommendations.	Minor omissions in documentation, but thorough analysis. Bottleneck identification is mostly accurate.	Inspection is basic, some performance metrics missing. Bottleneck analysis is incomplete.	Major errors in inspection or testing; no actionable recommendations.	No attempt or incomplete.



What to Check

Now that you've explored the contents, practice and activities of this lesson, it's time to reflect on what you've learned. Answer this short quiz

to assess your understanding on **Pre-Deployment procedures**. Apply what you've learned and carefully consider each question. Good luck!

 How to Access the Short Quiz:

 Click this link: <https://forms.gle/oGo5GSttC8658ib77>

OR

 Just kindly scan this QR Code to access the Short Quiz via Google Forms.

Scan this QR Code



What I have learned

Personal Reflection and Value Statement

Instruction:

Take a few minutes to reflect on your experiences related to teamwork and preparation in any context (academic, professional, or personal). Write a brief value statement (3-5 sentences) expressing your beliefs about the importance of pre-deployment practices, focusing on how these practices influence collaboration, team dynamics, and overall success

References:

Varde, D. (2024, January 22). *Deploying a Windows Server: A Comprehensive Guide*. LinkedIn. Retrieved from <https://www.linkedin.com/pulse/deploying-windows-server-comprehensive-guide-dhruv-varde-zrfgf>

Microsoft. (2024, August 11). *Set up and deploy on-premises environments (Application 10.0.32 and later)*. Microsoft Learn. Retrieved from <https://learn.microsoft.com/en-us/dynamics365/fin-ops-core/dev-itpro/deployment/setup-deploy-on-premises-latest>

Manifestly. (2014, June 1). *Server Setup Checklist*. Retrieved from <https://www.manifest.ly/use-cases/systems-administration/server-setup-checklist>

Microsoft. (2022, June 14). *Planning the Deployment*. Microsoft Learn. Retrieved from <https://learn.microsoft.com/en-us/iis/web-hosting/installing-infrastructure-components/planning-the-deployment>

OpenCRVS. (n.d.). *Pre-Deployment Checklist*. Retrieved from <https://documentation.opencrvs.org/setup/6.-go-live/3.3.4-set-up-an-smtp-server-for-opencrvs-monitoring-alerts>

TutorChase. (n.d.). *What are the risks associated with inadequate software deployment strategies?*. Retrieved from <https://www.tutorchase.com/answers/ib/computer-science/what-are-the-risks-associated-with-inadequate-software-deployment-strategies>

OpenMindT. (n.d.). *Software Deployment Guide: Methods, Risks & Best Practices*. Retrieved from <https://www.openmindt.com/knowledge/software-deployment-guide-definition-methods-risks/>